

EDUCATION

- **University of Illinois Urbana-Champaign** Urbana, IL, USA
Ph.D. of Computer Science; GPA 4.0/4.0 Aug. 2024 - May. 2029
 - **Research Field:** Stable and Autonomous RL Post-Training for Large Generative Models (Flow/Diffusion, Multi-modal Reasoning LLMs).
 - **Service:** Reviewer at ICML 2025–2026, ICLR 2025–2026, NeurIPS 2024–2026, COLM 2026, CVPR 2026, WACV 2027, AAAI 2025 & 2027, AISTATS 2025, KDD 2024.
 - **Relevant Course:** Machine Learning Algorithms for Large Language Models (LLMs).
- **Tsinghua University** Beijing, China
M.Eng. of Computer Technology; GPA 3.97/4.0, Top 1.3% Sept. 2021 - Jun. 2024
 - **Service:** Reviewer at NeurIPS 2022–2023, ICML 2023–2024, ICLR 2024.
 - **Relevant Course:** Stochastic Processes (A). Big Data Systems (A⁺). Digital Processing of Speech Signals (A). Data Visualization (A⁺).
- **Nankai University** Tianjin, China
B.Eng. of Intelligent Science and Technology; GPA 93.28/100 (3.9/4.0), 1/83 Sept. 2017 - Jun. 2021
 - **Honors:** Ranked 1st in major; National Scholarship twice (Top 1%).

RESEARCH INTEREST

- **Primary:** Stable and autonomous RL post-training for large generative models, spanning flow/diffusion models and multi-modal reasoning LLMs. Key focus on progressively reducing human intervention in the post-training pipeline—from online RL that eliminates human-collected data (ORW-CFM-W2), to adaptive divergence control for collapse-free training without manual KL tuning (ADRPO), to process rewards that remove reliance on human-annotated reasoning paths (CESAR), to fully autonomous self-critique where models become their own critics, eliminating hand-crafted rewards entirely.
- **Secondary:** Scalable and stable post-training methods that address critical challenges in RL-based fine-tuning: collapse-free training that preserves generation diversity throughout continuous optimization, adaptive exploration-exploitation trade-offs via self-regulated KL divergence, and effective test-time scaling that resolves inverse scaling in reasoning models—enabling superhuman-level audio reasoning capabilities.

PUBLICATIONS

1. Sun, J., Jiang, P., Wang, S., **Fan, J.**, Wang, H., Ouyang, S., Zhong, M., Jiao, Y., Huang, C., Xu, X., Han, P., Li, P., Huang, J., Liu, G., Ji, H., Han, J. Rethinking the Reranker: Boundary-Aware Evidence Selection for Robust Retrieval-Augmented Generation. International Conference on Machine Learning 2026 (**ICML 2026**).
2. Yang, B., Ma, H., **Fan, J.**, Liu, G. Batched Contextual Reinforcement. International Conference on Machine Learning 2026 (**ICML 2026**).
3. Zheng, Q.[†], **Fan, J.**[†], Cheng, C., Liu, G. Procedure-Aware Reinforcement Learning for Tool-Augmented Large Language Models. Conference on Language Modeling 2026 (**COLM 2026**), **co-first author**.
4. **Fan, J.**, Ren, R., Li, J., Pandey, R., Shivakumar, P.G., Gu, Y., Gandhe, A., Liu, G., Bulyko, I. Incentivizing Consistent, Effective and Scalable Reasoning Capability in Audio LLMs via Reasoning Process Rewards. International Conference on Learning Representations 2026 (**ICLR 2026**).
5. Y Li, Y Meng, Z Sun, K Ji, C Tang, **J Fan**, et al. SP-VLA: A Joint Model Scheduling and Token Pruning Approach for VLA Model Acceleration. International Conference on Learning Representations 2026 (**ICLR 2026**).
6. Wang, Z.[†], **Fan, J.**[†], et al. ProteinZero: Self-Improving Protein Generation via Online Reinforcement Learning. Transactions on Machine Learning Research (**TMLR**), 2026, **co-first author**.
7. Wang, Z., **Fan, J.**, et al. Variational Supervised Contrastive Learning. The Thirty-Ninth Annual Conference on Neural Information Processing Systems 2025 (**NeurIPS 2025**).
8. **Fan, J.**, et al. Adaptive Divergence Regularized Policy Optimization for Fine-tuning Generative Models. The Thirty-Ninth Annual Conference on Neural Information Processing Systems 2025 (**NeurIPS 2025**).

9. **Fan, J.**, et al. Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization. International Conference on Learning Representations 2025 (**ICLR 2025**).
10. Ye Li, Chen Tang, Yuan Meng, **Jiajun Fan**, et al. PRANCE: Joint Token-Optimization and Structural Channel-Pruning for Adaptive ViT Inference. IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2025.
11. **Fan, J.**, et al. Efficient Design-and-Control Automation with Reinforcement Learning and Adaptive Exploration. **NeurIPS 2024** Workshop AI4Mat.
12. **Fan, J.**, et al. Learnable Behavior Control: Breaking Atari Human World Records via Sample-Efficient Behavior Selection. International Conference on Learning Representations 2023 (**ICLR 2023**), **oral presentation, ranked 5/4176**.
13. Hao Wang, Chen Zhichao, **Jiajun Fan**, et al. Optimal Transport for Treatment Effect Estimation. The Conference on Neural Information Processing Systems 2023 (**NeurIPS 2023**).
14. **Fan, J.**, Xiao, C. Generalized Data Distribution Iteration. International Conference on Machine Learning 2022 (**ICML 2022**).
15. Xiao, C., Shi, H., **Fan, J.**, & Deng, S. CASA: A Bridge Between Gradient of Policy Improvement and Policy Evaluation. In the proceedings of Deep Reinforcement Learning Workshop **NeurIPS 2022**, 2022.
16. **Fan, J.** A Review for Deep Reinforcement Learning in Atari: Benchmarks, Challenges, and Solutions. In the proceedings of **AAAI-22** Workshop on Reinforcement Learning in Games, 2021.
17. **Fan, J.**, Xiao, C., & Huang, Y. GDI: Rethinking What Makes Reinforcement Learning Different From Supervised Learning. In the proceedings of **AAAI-22** Workshop on Reinforcement Learning in Games, 2021.
18. Wang E., Lian J., **Fan J.**, et al. Enhancing Sequential User Modeling with Large-kernel Convolution: A Lightweight Approach. The 30th SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2024, under review.
19. Wang E., Can Z., Yang Y., Pan L., **Fan J.**, et al. Unbiased Recommender Learning from Implicit Feedback: A Weak Supervision Perspective. The 30th SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2024, under review.
20. Wang E., **Fan J.**, et al. SPEC: Constructing Reliable Sequential User Model using Slide-window Spectrum. ACM TheWebConf 2024 Conference, under review.
21. E. Wang, H. Li, T. Liu, Y. Yang, **Fan J.**, X. Liu, and Z. Chen, “Unbiased recommender learning from implicit feedback: A progressive proximal transport approach,” in ACM TheWebConf 2024 Conference, under review.
22. Wang, H., Chen, Z., **Fan, J.**, et al. Entire Space Counterfactual Learning: Tuning, Analytical Properties and Industrial Applications. The IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**), under review.
23. Xiao, C., Shi, H., **Fan, J.**, & Deng, S. An Entropy Regularization Free Mechanism for Policy-based Reinforcement Learning. arXiv preprint arXiv:2106.00707.
24. **Fan, J.**, Ba, H., Guo, X., & Hao, J. Critic PI2: Master Continuous Planning via Policy Improvement with Path Integrals and Deep Actor-Critic Reinforcement Learning. arXiv preprint arXiv:2011.06752.

INDUSTRY EXPERIENCE

- **Amazon — AGI Foundations (Audio / Multi-modal LLM)** Seattle, WA, USA
Applied Scientist Intern (Returning Intern) May. 2026 - Present
 - **Scope:** Invited back as a returning intern; ongoing work on full-stack distributed agentic RL training. Details pending publication.
- **Amazon — AGI Foundations (Audio / Multi-modal LLM)** Seattle, WA, USA
Applied Scientist Intern May. 2025 - Dec. 2025
 - **Full-Stack Distributed RL Post-Training:** Sole owner of the **end-to-end pipeline** for multi-modal reasoning LLMs—distributed training infrastructure, algorithm and reward design, evaluation, and model delivery—and sole first author of the resulting publication.
 - **Infrastructure & Algorithm:** Built and scaled a **multi-node, multi-GPU GRPO training stack** for large-scale omni-modal LLMs, and designed **CESAR**, a process-reward framework that moves RLVR beyond outcome verification and removes any dependence on human-annotated reasoning traces.

- **Impact:** Diagnosed and resolved test-time inverse scaling—the failure mode that had made chain-of-thought counterproductive in Audio LLMs—and delivered state-of-the-art results on MMAU Test-mini, surpassing Gemini 2.5 Pro and GPT-4o Audio, with near-human-level reasoning on MMSU. Published at **ICLR 2026** as first author.

RESEARCH HIGHLIGHTS

• Self-Evolving RLHF for Flow Matching Generative Models

Sept. 2024 - Present

- **Role:** **1)** Introduced a self-evolving RLHF framework (ORW-CFM-W2) that enables flow matching models to continuously optimize through online reward feedback without relying on human-collected datasets or likelihood calculations. **2)** Derived a tractable Wasserstein-2 distance bound for flow matching models, providing the first theoretical guarantee for collapse-free policy evolution. **3)** Established a unified perspective connecting flow matching fine-tuning with traditional KL-regularized RL, enabling controllable reward-diversity trade-offs.
- **Achievements:** **1)** Achieved state-of-the-art alignment with orders of magnitude less data while maintaining generation diversity through theoretically-grounded regularization. **2)** Validated the framework's effectiveness by successfully fine-tuning large-scale models like Stable Diffusion 3 across challenging tasks including spatial understanding and compositional generation. **3)** Published a paper called "Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization" at **ICLR 2025** as first author.

• Behavior Control in Reinforcement Learning

Sept. 2021 - Aug. 2024

- **Role:** **1)** Introduced a unified framework called LBC to achieve behavior control in RL. **2)** Provided a unified perspective on diverse RL methods for behavior control and potential enhancements. **3)** Validated LBC's efficacy through rigorous theoretical support and extensive empirical experiments.
- **Achievements:** **1)** Surpassed 24 human world records and attained the pinnacle of performance among reinforcement learning algorithms across most tasks. **2)** Published a paper titled "Learnable Behavior Control: Eclipsing Human World Records in Atari Games through Sample-Efficient Behavior Selection" at **ICLR 2023** with oral presentation as first author.

• Sample-Efficient Reinforcement Learning

Sept. 2020 - Aug. 2024

- **Role:** **1)** Introduced a sample-efficient Reinforcement Learning (RL) framework known as GDI, which achieved human-level performance by optimizing the data distribution of RL agents. **2)** Supported the efficacy of GDI with a robust foundation, including both theoretical proofs and an extensive array of experiments conducted in Atari. **3)** Provided a unified perspective on various RL algorithms with GDI.
- **Achievements:** **1)** Outperformed prior SOTA method Agent57 with 500x less data and twice the average performance. **2)** Published a paper called "Generalized Data Distribution Iteration" at **ICML 2022** as first author.

PATENTS

- Unified framework for model-free reinforcement learning algorithms Fan, J., Xiao, C. Unified framework for model-free reinforcement learning algorithms. CN112766497A[P].
- Hyperparameter tuning algorithm based on multi-arm gambling machine optimizer Fan, J. Hyperparameter tuning algorithm based on multi-arm gambling machine optimizer. CN112926629A[P].
- An unbiased estimation algorithm of behavior value function Fan, J., Xiao, C. An unbiased estimation algorithm of behavior value function. CN112926628A[P].
- Policy gradient algorithm based on double robust qualification trace Fan, J., Xiao, C. Policy gradient algorithm based on double robust qualification trace. CN112926735A[P].
- Asynchronous multi-arm gambling machine hyperparameter optimizer based on electoral college voting mechanism Fan, J. Asynchronous multi-arm gambling machine hyperparameter optimizer based on electoral college voting mechanism. CN112949850A[P].
- Real-time multi-hyperparameter controller Fan, J. Real-time multi-hyperparameter controller. CN113052252A[P].
- Hyperspace multi-coupling parameter optimizer based on multi-arm gambling machine combined with democratic voting Fan, J. Hyperspace multi-coupling parameter optimizer based on multi-arm gambling machine combined with democratic voting. CN113052253A[P].

- Fast and generalizable hyperspace coupling multi-parameter nonlinear optimizer Fan, J. Fast and generalizable hyperspace coupling multi-parameter nonlinear optimizer. CN113052248A[P].
- Reinforcement learning algorithm based on generalized combination strategy space Fan, J. Reinforcement learning algorithm based on generalized combination strategy space. CN113052312A[P].

AWARDS

- Outstanding Graduates (1%) Tianjin, China, 2021
- Excellent Graduation Thesis of Nankai University Tianjin, China, 2021
- Tang Lixin Scholarship (1%) Tianjin, China, 2021
- **National Scholarship**, Nankai University (1/83) Tianjin, China, 2020
- Nomination for Zhou Enlai Scholarship Tianjin, China, 2020
- **National Scholarship**, Nankai University (1/83) Tianjin, China, 2019
- 3rd Prize, Robocup@HOME Education World Final Sydney, Australia, 2019
- Bronze Medal, **ACM / ICPC** Asia Regional Contest Xuzhou, China, 2019
- National 2nd Prize, National College Students Mathematical Contest in Modeling (5%) Tianjin, China, 2018
- The First Prize Scholarship, Nankai University (2/83) Tianjin, China, 2018

RESEARCH EXPERIENCE

- **RL Post-Training for Audio Large Language Models** Ph.D. Student
UIUC, Urbana, IL, USA May. 2025 - Present
 - **Motivation:** Reasoning in Audio LLMs remains underexplored. Chain-of-thought reasoning often degrades inference performance—a phenomenon we term *test-time inverse scaling*—due to hallucinatory and inconsistent reasoning. Existing RL methods rely on outcome-only rewards or require laborious hand-crafted process rewards, failing to cultivate genuine reasoning capability.
 - **Results:** Proposed CESAR, an online RL framework using GRPO with multi-faceted reasoning process rewards that incentivize consistency, structured reasoning, and calibrated depth. Resolved test-time inverse scaling and achieved SOTA on MMAU Test-mini, outperforming Gemini 2.5 Pro and GPT-4o Audio.
 - **Publication:** “Incentivizing Consistent, Effective and Scalable Reasoning Capability in Audio LLMs via Reasoning Process Rewards” at **ICLR 2026** as first author.
- **Procedure-Aware RL for Tool-Augmented / Agentic LLMs** Ph.D. Student
UIUC, Urbana, IL, USA Oct. 2025 - Present
 - **Motivation:** Existing RLVR methods for tool use reward the correct *set* of tool calls while ignoring their execution order, producing agents that are fragile in state-sensitive, multi-turn environments.
 - **Results:** Introduced a procedure-aware reward decomposition (NDCG-based trace reward, suffix-matching state reward, and structural error penalties) under GRPO. On BFCL-V3 the 3B model reaches performance competitive with proprietary models, with a **+87% relative multi-turn gain** over the correctness-only RL baseline, and consistent gains on API-Bank and Bamboogle.
 - **Publication:** “Procedure-Aware Reinforcement Learning for Tool-Augmented Large Language Models” at **COLM 2026** as **co-first author**.
- **RL Post-Training for Flow/Diffusion Generative Models** Ph.D. Student
UIUC, Urbana, IL, USA Aug. 2024 - May. 2025
 - **Motivation:** Large generative models like diffusion and flow matching models lack the ability to continuously improve from reward feedback without catastrophic forgetting or mode collapse. Existing fine-tuning methods either require expensive human-collected datasets or fail to maintain generation diversity.
 - **Results:** Achieved state-of-the-art alignment performance with orders of magnitude less data. Successfully fine-tuned large-scale models including Stable Diffusion 3 on challenging tasks such as spatial understanding and compositional generation, while preserving generation diversity.
 - **Publication:** Published “Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization” at **ICLR 2025** as first author. Published “Adaptive Divergence Regularized Policy Optimization for Fine-tuning Generative Models” at **NeurIPS 2025** as first author.

- **Reinforcement Learning for Robotics** Research Assistant
Oct. 2023 - May. 2024
Mila, Montreal, Canada
 - **Motivation:** Robot design and control are typically treated as separate problems, leading to suboptimal solutions. Jointly optimizing both in an end-to-end manner can significantly improve overall system performance.
 - **Results:** Combined the strengths of model-free RL and design-based RL, improving multiple metrics in MuJoCo control tasks. Introduced behavior control into design-based RL, improving sample efficiency.
 - **Publication:** Published at NeurIPS 2024 Workshop AI4Mat.
- **Behavior Control in Reinforcement Learning** Research Assistant
Jun. 2021 - Oct. 2023
Shenzhen, China
 - **Motivation:** Existing RL algorithms struggle with complex tasks requiring diverse exploration strategies. A unified framework for controlling agent behavior through adaptive strategy selection was needed.
 - **Results:** Surpassed **24 human world records** in Atari games and achieved top performance among RL algorithms across most tasks, demonstrating exceptional sample efficiency.
 - **Publication:** Published “Learnable Behavior Control: Breaking Atari Human World Records via Sample-Efficient Behavior Selection” at ICLR 2023 with oral presentation (ranked 5/4176), as first author.
- **Sample-Efficient Reinforcement Learning (GDI)** Research Assistant
Sep. 2020 - Jun. 2021
Beijing, China
 - **Motivation:** State-of-the-art RL methods like Agent57 required enormous amounts of data to achieve human-level performance in Atari games. A more data-efficient paradigm was needed to make RL practical for real-world applications.
 - **Results:** Outperformed Agent57 with **500x less data** and **twice** the average performance on the Atari-57 benchmark, surpassing **22 human world records**.
 - **Publication:** Published “Generalized Data Distribution Iteration” at ICML 2022 as first author.
- **Model-based Reinforcement Learning Algorithm** Research Assistant
Apr. 2020 - Sep. 2020
Tianjin, China
 - **Role:** Introduced a groundbreaking method for achieving unparalleled performance on the challenging MuJoCo environment, which represents a significant breakthrough in the field of reinforcement learning.
 - **Achievements:** Presented a cutting-edge paper titled “Critic PI2: Master Continuous Planning via Policy Improvement with Path Integrals and Deep Actor-Critic Reinforcement Learning.”
- **Multifunctional Home Service Robot** Research Assistant
Sep. 2018 - Jul. 2019
Tianjin, China
 - **Role:** **1)** Devised a comprehensive solution for the entire project, along with a finite state machine diagram. **2)** Employed cutting-edge ROS-based algorithms for autonomous navigation and RRT path planning.
 - **Achievements:** Presented the project in the 2019 ROBOCUP Sydney World Finals and won **3rd Prize**.

ACADEMIC SERVICE

- Reviewer of The 41st Annual AAAI Conference on Artificial Intelligence AAAI 2027
- Reviewer of The IEEE/CVF Winter Conference on Applications of Computer Vision WACV 2027
- Reviewer of The Fortieth Annual Conference on Neural Information Processing Systems NeurIPS 2026
- Reviewer of The Second Conference on Language Modeling COLM 2026
- Reviewer of The Forty-Third International Conference on Machine Learning ICML 2026
- Reviewer of The Fourteenth International Conference on Learning Representations ICLR 2026
- Reviewer of The IEEE/CVF Conference on Computer Vision and Pattern Recognition CVPR 2026
- Reviewer of Transactions on Machine Learning Research TMLR
- Reviewer of The 28th International Conference on Artificial Intelligence and Statistics AISTATS 2025
- Reviewer of Forty-Second International Conference on Machine Learning ICML 2025
- Reviewer of The Thirteenth International Conference on Learning Representations ICLR 2025
- Reviewer of The 39th Annual AAAI Conference on Artificial Intelligence AAAI 2025

- Reviewer of The Thirty-Eighth Annual Conference on Neural Information Processing Systems NeurIPS 2024
- Reviewer of The 30th SIGKDD Conference on Knowledge Discovery and Data Mining KDD 2024
- Reviewer of The Forty-first International Conference on Machine Learning ICML 2024
- Reviewer of The Twelfth International Conference on Learning Representations ICLR 2024
- Reviewer of The Thirty-seventh Annual Conference on Neural Information Processing Systems. NeurIPS 2023
- Reviewer of The Fortieth International Conference on Machine Learning. ICML 2023
- Reviewer of The Thirty-sixth Annual Conference on Neural Information Processing Systems. NeurIPS 2022